Galaxy Genome Annotation

Galaxy as a platform for the annotation of genomes

Anthony Bretaudeau[1,2], Helena Rasche[3] - https://galaxy-genome-annotation.github.io

Community effort to ease genome annotation with Galaxy
Come chat with us!
https://matrix.to/#/#galaxy-genome-annotation_Lobby:gitter.im

A full catalog of tools

- Gene prediction: **Prokka**, **Bakta**, **Maker**, **Braker2**, **Braker3**, **Helixer**, **Funannotate**, **Augustus**, **SNAP**
- Functional annotation : **Antismash**, **InterProScan**, **EggNOG-Mapper**
- Quality assessment: **BUSCO**
- Visualisation: **JBrowse**, **Circos**, **GeneNoteBook**
- Repeated elements: **Red**, **RepeatMasker**, **RepeatModeler**
- Long non-coding RNAs: **FEELnc**

Training Material

Training material on the **Galaxy Training Network**

- Annotation of prokaryotes with Prokka
- Annotation of eukaryotes with Maker or Funannotate
- Functional annotation
- Repeat detection
- Long non-coding RNA detection
-

https://training.galaxyproject.org - Genome annotation section

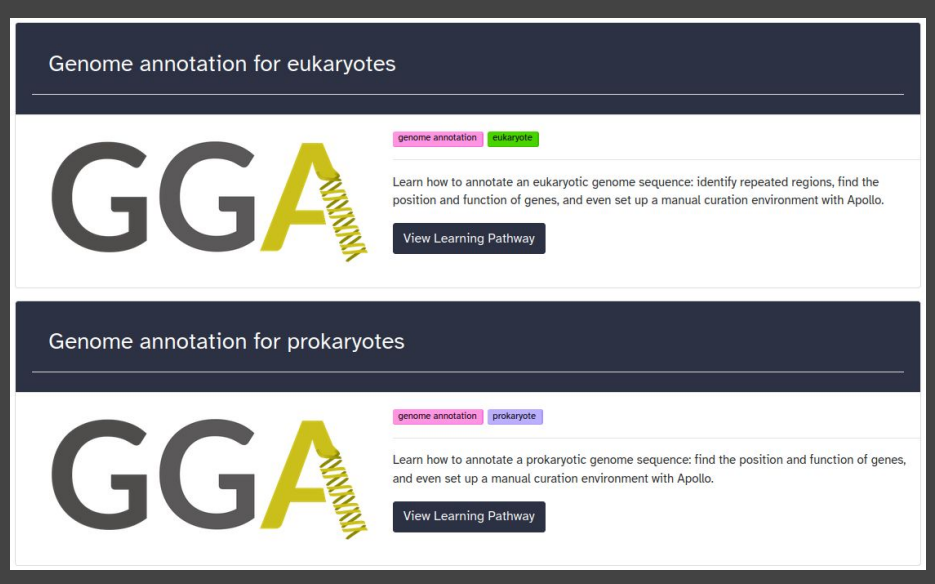
Gallantries

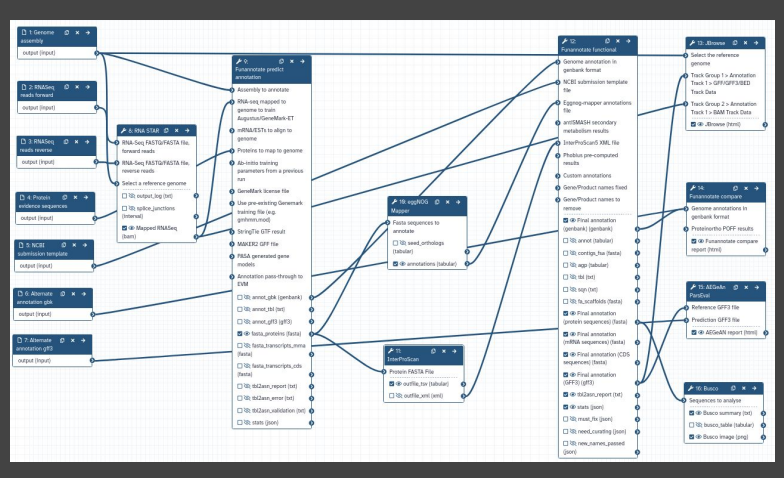
Contributions to the Erasmus+ Gallantries project (2020-2023)

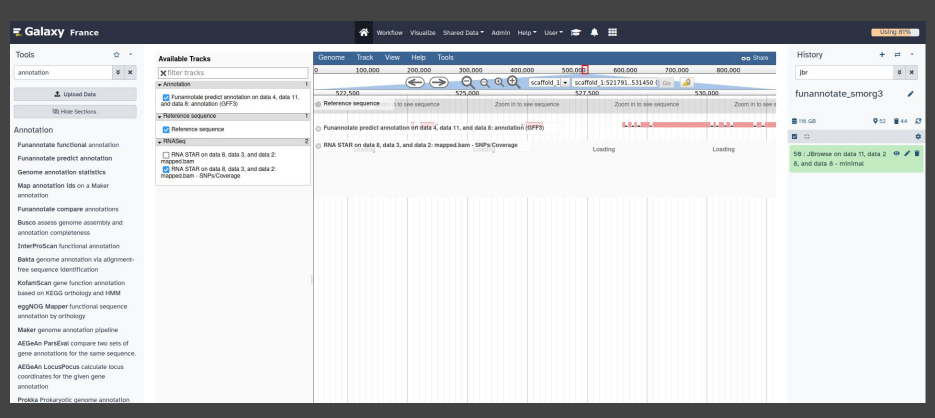
- GTN content (tutorial, workflows and videos)
- Full learning pathways
- Smörgåsbord events

Smörgåsbord 2023 :

- 2063 / 3321 participants interested
- 3000 page views
- 250 videos views







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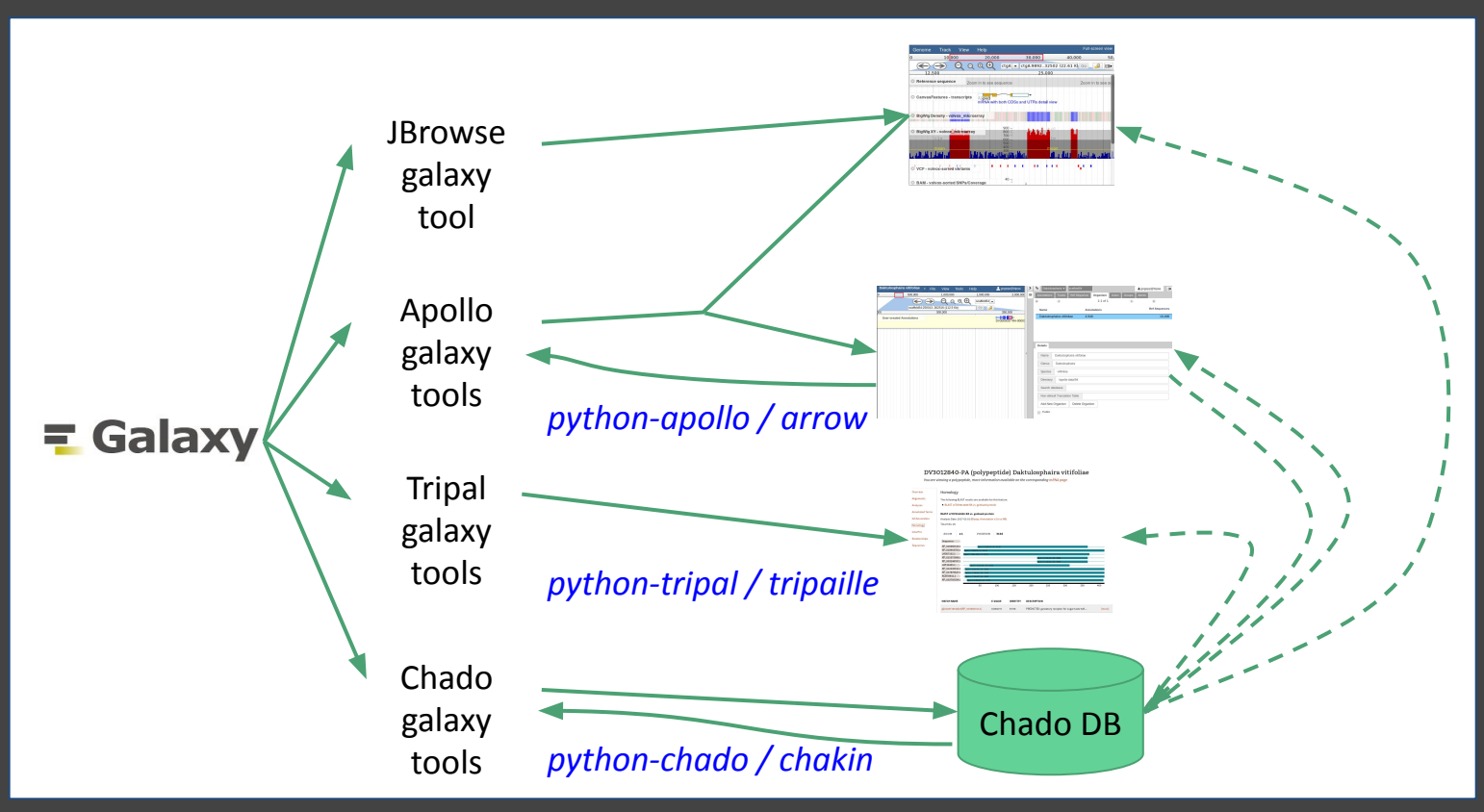
GMOD Integration

- **JBrowse**: web-based genome browser
- **Apollo**: web-based genome annotation editor
- **Chado**: PostgreSQL database schema for storing biological data
- **Tripal**: Drupal-based application for creating biological websites

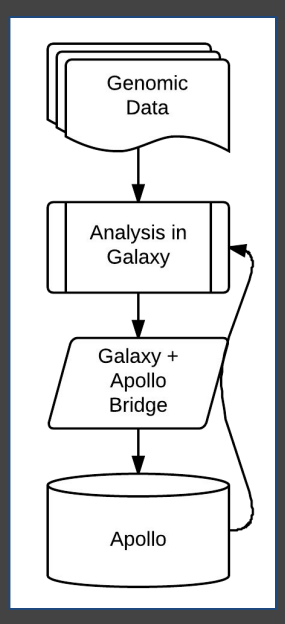
GGA maintains:

- **Docker images** for GMOD applications (JBrowse, Apollo, Tripal, Chado)
- **Python libraries**: python-chado, python-apollo, python-tripal
- **Command line tools**: chakin (chado), arrow (apollo), tripaille (tripal)
- **Galaxy tools** to interact with GMOD apps: load, and/or extract data



Manual curation with Apollo

- Complete workflow with Galaxy:
- Predict genes on your newly assembled genome
- Visualise the annotation by using the JBrowse Galaxy tool
- Transfer the JBrowse dataset to an Apollo server
- Manage access permissions
- Multi-user manual curation on the Apollo server
- Analysis/release of the improved annotation



Apollo available on **https://usegalaxy.eu/apollo**

Relies on:

Apolpi - optimized Apollo API
https://github.com/galaxy-genome-annotation/apolpi
Permapol - permission management UI
https://github.com/genouest/permapol/

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Genome Portals

GGA resources can be used for the creation of genome portals

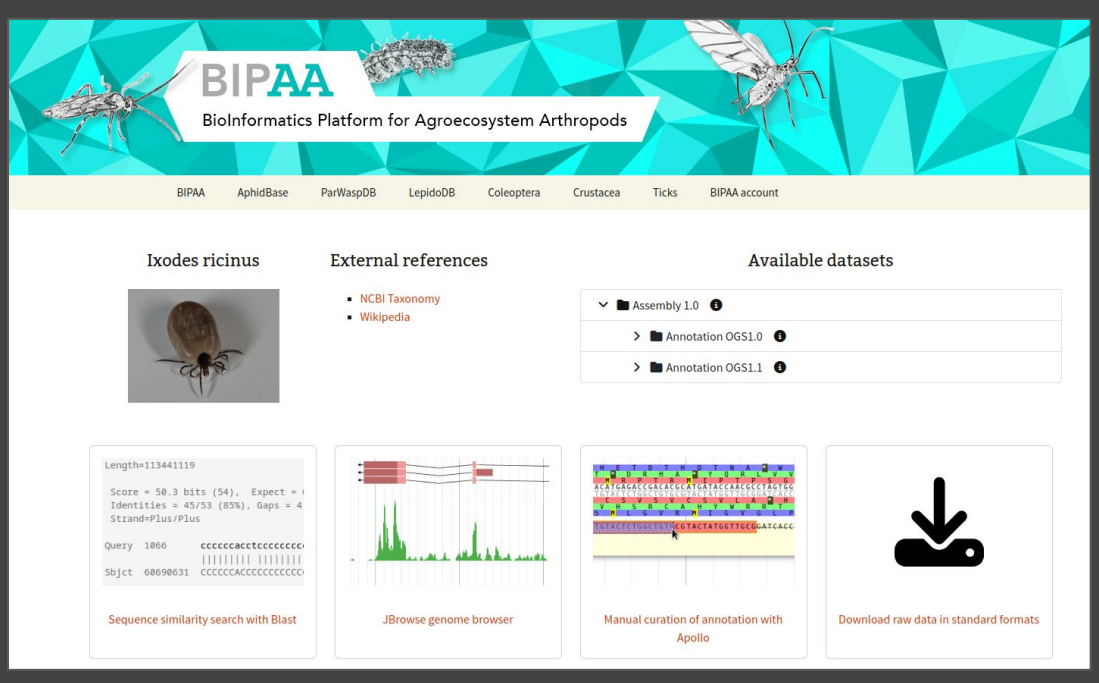
1 genome = 1 set of dockerized web applications using Docker-compose

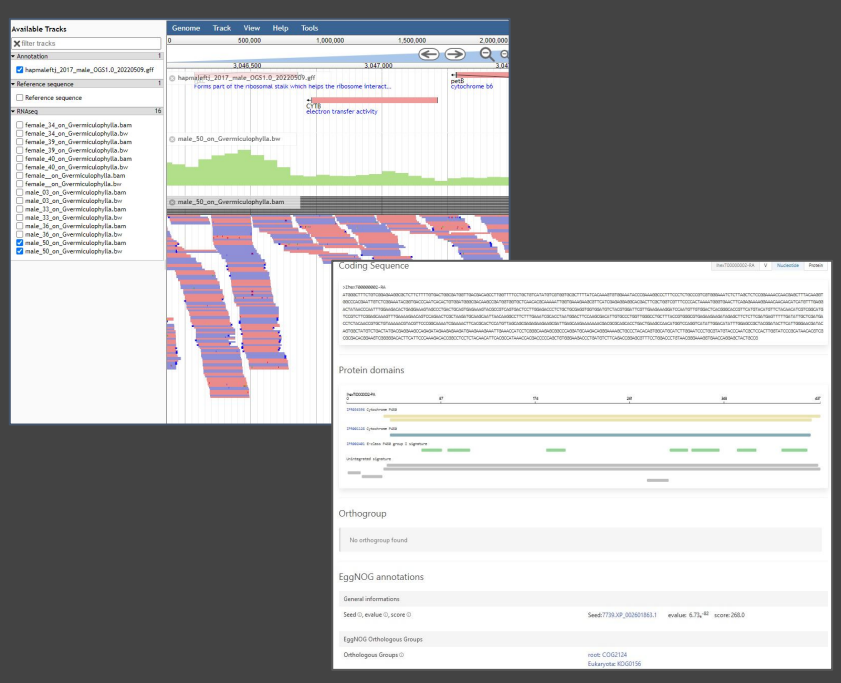
Implemented on two databases:

- BIPAA (insect genomes) - https://bipaa.genouest.org
- Algae - https://abims.sb-roscoff.fr/resources/genomic_resources

BEAURIS: new project for full automation of genomic web portal creation

BEAURIS uses GGA internally - https://gitlab.com/beaur1s/beauris





Ongoing projects

EuroScienceGateway

European network of computing resources (**Pulsar**) and national usegalaxy.*

Biodiversity use case with genome annotation workflows

Big sequencing projects: VGP/ERGA/BGE/ATLASea

Thousands of genome sequenced in the coming years

Need for FAIR standard annotation workflows

Implementing Galaxy workflows, in the same way as VGP assembly workflows

Building genomic portal with GGA components (ATLASea)

Thanks!

- Galaxy / GMOD / Elixir / ERGA / VGP communities
- UseGalaxy.* / GTN / IFB (Elixir-FR) infrastructures
- UseGalaxy.fr team: Gildas Le Corguillé, Thomas Chaussepied, Romane Libouban, Eva Mercier, Olivier Collin, …
- GenOuest, BIPAA, ABiMS

1. IGEPP, INRAE, Institut Agro, Univ Rennes, Rennes, France
2. GenOuest Core Facility, Univ Rennes, Inria, CNRS, IRISA, Rennes - abretaud - abretau - anthony.bretaudeau@inrae.fr
3. Clinical Bioinformatics Group, Department of Pathology, Erasmus Medical Center, Rotterdam, the Netherlands

All code contributed to GGA/GMOD/Galaxy github repositories under open-source license

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